Online principal-subspace estimation with adaptive Markov chain Monte Carlo

A controlled stochastic approximation analysis

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Abstract

We investigate whether Oja's online principal-subspace iteration can be coupled to adaptive Markov chain Monte Carlo without assuming independent samples. The adaptive mean and covariance estimates follow the recursion considered by Andrieu and Moulines, while an additional Stiefel-valued iterate is updated through a polar retraction. We identify the common averaged vector field, establish a uniform second-order bound for the retraction error, and give an explicit weighted Poisson decomposition of the controlled Markov fluctuations. A localized convergence theorem shows that, on eventually compact parameter trajectories, the estimates approach the target moments and the Oja iterate approaches the collection of invariant subspaces of the target covariance. A composite Lyapunov function based on the expected PCA reconstruction error gives an exact global dissipation identity for the averaged coupled system. The remaining global stability question, needed for an unconditional convergence theorem for the untruncated algorithm, is isolated explicitly.

1 Introduction

Online principal component analysis (PCA) is often presented as a stochastic approximation algorithm driven by independent observations. In a Monte Carlo application, however, observations are naturally successive states of a Markov chain. Furthermore, when the sampler adapts its proposal distribution, even the transition kernel changes over time. The central question of this note is consequently concrete:

Can an Oja principal-subspace recursion be run on the output of an adaptive MCMC sampler while retaining a rigorous stochastic approximation interpretation and convergence guarantee?

We consider a target distribution π on $\mathbb{R}^d$ and a Gaussian random-walk Metropolis family with proposal covariance updated online. The mean–covariance adaptation is that of [1]. The addition is an orthonormal frame W_n whose columns are updated from the **same accepted MCMC states** through a projected Oja increment and a Stiefel retraction. This separates two issues often conflated in informal discussions of PCA adaptation: the estimation of the target covariance and the choice of a low-dimensional invariant subspace of that covariance.

Three mathematical observations make the coupling tractable. First, all frozen MCMC kernels have the same invariant distribution π. Their time-inhomogeneity therefore affects the **noise averaging argument** but not the algebraic form of the mean field. Second, a polar retraction can be expanded in the ambient matrix space with a quadratic remainder, allowing Euclidean stochastic approximation estimates without changing the geometry of the actual iterates. Third, the dependence of both the covariance and Oja updates on the

evolving mean identifies a natural expected reconstruction-error objective. Together with the mean–covariance Lyapunov function, it gives an exact global dissipation identity for the coupled averaged flow.

The paper follows the proof rather than concealing its mechanics. We derive the mean field, state weighted-kernel assumptions, expand the retraction, show the exact telescoping terms in the Poisson decomposition, and differentiate the Lyapunov function explicitly. The initial convergence conclusion is deliberately the approach to the set of **invariant r-dimensional subspaces**. Almost-sure selection of the leading eigenspace is a stronger claim and is not implied by deterministic eigenvalue gaps alone.

1.1 Relation to existing results

Andrieu and Moulines establish a general controlled-Markov framework for adaptive MCMC, including kernel drift and regularity estimates, Poisson equations, and a random-truncation stability method [1]. We retain that framework and its condition (M), rather than assuming independent samples. The retraction calculations use standard embedded-manifold Taylor expansions as developed by [2]. The expected reconstruction-error objective follows the classical Lyapunov treatment of principal-component algorithms [3]. We use the ordinary differential equation (ODE) method for stochastic approximation and the associated internally chain-transitive limit-set principle [4].

The contribution is a transparent synthesis for this particular coupled recursion, not a new general theorem about manifold-valued Markov processes. In particular, the first main theorem is **localized**: it proves convergence on sample paths whose mean–covariance estimates ultimately remain in a compact subset of the positive-definite parameter domain. The finite-reinitialization step required to remove that qualification for the enlarged adaptive algorithm is listed as a separate proof obligation.

2 Notation and conventions

This page fixes notation used throughout the manuscript. The conventions for weighted functions, the Markov kernel and Poisson equations follow [1]; the notation for the Stiefel manifold and retractions follows [2].

Symbol	Meaning
d, r	State dimension and subspace dimension, $1 \le r < d$
X_n	MCMC state after the n-th transition
π	Common invariant probability distribution of the frozen kernels
m, C	Mean and covariance of π: $m = \pi(x)$, $C = \pi[(x-m)(x-m)^\top]$
μ_n, Γ_n	Adaptive estimates of m and C
$\mathbb{S}^d_{++}$	Real symmetric, **strictly** positive-definite $d \times d$ matrices
$\mathrm{St}(d,r)$	$\{W \in \mathbb{R}^{d\times r} : W^\top W = I_r\}$
W_n	Stiefel iterate; $\mathrm{span}(W_n)$ is its subspace
P_W	Orthogonal residual projector $I - WW^\top$
$\mathcal{R}(\mu, W)$	Expected centred PCA reconstruction error $\frac{1}{2}\pi[\|P_W(X-\mu)\|^2]$
$\mathcal{L}_\lambda$	Joint Lyapunov function $w_{\mathrm{AM}} + \lambda\mathcal{R}$, $\lambda > 0$
θ_n	Full parameter (μ_n, Γ_n, W_n)
P_θ	Frozen MCMC kernel; in the example $P_\theta = P_\Gamma$
$H(\theta, x)$	Stochastic approximation observable in ambient coordinates
$h(\theta)$	Mean field $\pi[H(\theta,\cdot)]$
γ_n	Positive decreasing stochastic approximation step size
$R_W(U)$	Polar retraction of tangent increment U at W
$V : \mathbb{R}^d \to [1,\infty)$	Markov drift/weight function
$\|f\|_{V^a}$	$\sup_x \|f(x)\|/V^a(x)$, for scalar, vector or matrix f
u_θ	Centered Poisson solution $u_\theta - P_\theta u_\theta = H_\theta - h(\theta)$
v_θ	$P_\theta u_\theta$, used in summation by parts
$\mathcal{E}_C$	$\{W \in \mathrm{St}(d,r) : (I - WW^\top)CW = 0\}$

Our vectors use the Euclidean norm, matrices the Frobenius norm $\|\cdot\|_F$, and operators the spectral norm $\|\cdot\|_{\mathrm{op}}$. Products of gradients and vector fields use the product Euclidean–Frobenius inner product. We abbreviate integration against π as $\pi(f) = \int f(x)\pi(dx)$.

Throughout, $C \succ 0$ refers to the target covariance, whereas $\Gamma \succ 0$ is the time-varying estimate. The letter V is **reserved for the Markov drift function**, never for the covariance. We also write

$$v = \mu - m, \qquad Q(\mu) = C + vv^\top, \qquad B_C(W) = (I - WW^\top)CW. \tag{2.1}$$

All stochastic convergence statements specify whether they are unconditional or asserted only on an eventual compactness event. The term *equilibrium set* refers to equilibria of the averaged ODE, not the set of roots of an arbitrary projected noisy step.

3 The coupled algorithm and its mean field

3.1 State space and MCMC transition

Fix π on $\mathbb{R}^d$, with finite mean m and strictly positive-definite covariance C. Let

$$\Theta = \mathbb{R}^d \times \mathbb{S}^d_{++} \times \mathrm{St}(d,r), \qquad \theta = (\mu, \Gamma, W). \tag{3.1}$$

For definiteness, P_Γ is the symmetric random-walk Metropolis kernel targeting π, with proposal $Y = x + \varepsilon$, $\varepsilon \sim N(0, \ell^2\Gamma)$, for fixed $\ell > 0$. In particular $P_\theta = P_\Gamma$ does not depend on μ or W, and

$$\pi P_\Gamma = \pi, \qquad \Gamma \in \mathbb{S}^d_{++}. \tag{3.2}$$

A possible implementation may allow further dependence on W, but that would require a separate kernel-regularity verification. We do not assume it here.

3.2 Sample-driven Oja iteration

Define the horizontal Oja increment

$$G(\mu, W, x) = (I - WW^\top)(x-\mu)(x-\mu)^\top W. \tag{3.3}$$

Since $W^\top G(\mu, W, x) = 0$, this is tangent to the Stiefel manifold. We use the polar retraction

$$R_W(U) = (W+U)(I_r + U^\top U)^{-1/2}, \qquad W^\top U = 0. \tag{3.4}$$

Definition 3.1 (Coupled adaptive Markov–Oja recursion)**.** Given $X_n, \mu_n, \Gamma_n, W_n$ with $\Gamma_n \succ 0$ and $W_n^\top W_n = I_r$, draw $X_{n+1} \sim P_{\Gamma_n}(X_n, \cdot)$ and update

$$\begin{aligned} \mu_{n+1} &= \mu_n + \gamma_{n+1}(X_{n+1} - \mu_n), \\ \Gamma_{n+1} &= \Gamma_n + \gamma_{n+1}\big[(X_{n+1}-\mu_n)(X_{n+1}-\mu_n)^\top - \Gamma_n\big], \\ W_{n+1} &= R_{W_n}\big(\gamma_{n+1} G(\mu_n, W_n, X_{n+1})\big)\,. \end{aligned} \tag{3.5}$$

We take $0 < \gamma_n < 1$ to preserve $\Gamma_n \succ 0$ by convexity.

The first two lines are the mean–covariance adaptation of [1], equations (4)–(6). The third line is an additional Oja estimator based on the *same* accepted states. There is no independently simulated PCA data stream.

3.3 Averaging over the common invariant distribution

For $z = x - \mu$, define the ambient observable

$$H(\theta, x) = \begin{pmatrix} x - \mu \\ (x-\mu)(x-\mu)^\top - \Gamma \\ G(\mu, W, x) \end{pmatrix}. \tag{3.6}$$

The identity $x - \mu = (x - m) - (\mu - m)$, followed by expansion of the outer product, gives

$$\pi\big[(x-\mu)(x-\mu)^\top\big] = C + (\mu - m)(\mu - m)^\top = Q(\mu). \tag{3.7}$$

The mixed terms vanish because $\pi(x - m) = 0$. Therefore

$$h(\theta) = \pi[H(\theta, \cdot)] = \begin{pmatrix} m - \mu \\ Q(\mu) - \Gamma \\ (I - WW^\top)Q(\mu)W \end{pmatrix}. \tag{3.8}$$

This averaged field is the same algebraically whether the innovations are independent with law π or are generated by the changing kernels P_{Γ_n}. What changes is the justification of replacing path averages by π-averages.

The limiting ODE is consequently

$$\dot{\mu} = m - \mu, \quad \dot{\Gamma} = Q(\mu) - \Gamma, \quad \dot{W} = (I - WW^\top)Q(\mu)W. \tag{3.9}$$

The W-field is horizontal: $W^\top \dot{W} = 0$, hence $\frac{\mathrm{d}}{\mathrm{d}t}(W^\top W) = 0$ for the embedded-space ODE. The Stiefel manifold is invariant under its flow. The covariance-driven variant $\dot{W} = (I - WW^\top)\Gamma W$ is a **different algorithm** and is not used in the main theorem.

4 Markov assumptions and the Poisson equation

The central obstacle is not the algebra of PCA but the fact that the same Markov chain is used to update its own transition kernel. The estimates below use weighted norms in the sense of [1], Section 3, and its common-invariant-distribution framework.

4.1 Target and kernel conditions

Assumption 4.1 (Andrieu–Moulines condition (M))**.** *The target density π on $\mathbb{R}^d$ is bounded, continuously differentiable and bounded away from zero on every compact subset. It satisfies*

$$\lim_{\|x\| \to \infty} \left\langle \frac{x}{\|x\|}, \nabla \log \pi(x) \right\rangle = -\infty \tag{4.1}$$

and

$$\limsup_{\|x\| \to \infty} \left\langle \frac{x}{\|x\|}, \frac{\nabla \pi(x)}{\|\nabla \pi(x)\|} \right\rangle < 0. \tag{4.2}$$

These super-exponential tail and contour conditions are sufficient, not necessary, conditions for the random-walk Metropolis estimates used here. They imply that all polynomial moments of π exist; the strictly positive density implies the target covariance C is positive definite. Following A&M, choose

$$V(x) = \pi(x)^{-\eta}/(\sup_y \pi(y))^{-\eta}, \qquad 0 < \eta < 1, \tag{4.3}$$

so $V \geq 1$. For a vector- or matrix-valued observable f, use $\|f\|_{V^a} = \sup_x \|f(x)\|/V^a(x)$.

Assumption 4.2 (Locally uniform Markov kernel control)**.** *For every compact $K \subset \mathbb{R}^d \times \mathbb{S}^d_{++}$, the family P_Γ, $(\mu, \Gamma) \in K$, satisfies uniform minorization and geometric drift with weight V, and the weighted operator bound*

$$\|(P_\Gamma - P_{\Gamma'})f\|_{V^a} \leq L_{K,a}\|\Gamma - \Gamma'\|_F \|f\|_{V^a}, \qquad 0 \leq a \leq 1. \tag{4.4}$$

Every P_Γ preserves the same π.

Proposition 12 and Lemma 13 of [1] establish precisely these ingredients for the symmetric Gaussian random-walk Metropolis family under Assumption 4.1. Notice that $P_{\mu,\Gamma,W} = P_\Gamma$: adjoining W adds **no new kernel dependence**. Compactness of $\mathrm{St}(d, r)$ makes the constants uniform in W.

For a chain stopped when (μ_n, Γ_n) exits a compact parameter region, the drift bound yields a localized moment estimate of the form

$$\sup_{n \geq 0} \mathbb{E}[V(X_n)\mathbf{1}_{\{\tau_K \geq n\}}] < \infty. \tag{4.5}$$

This is the input used repeatedly in the summation arguments.

Assumption 4.3 (Step sizes)**.** *The deterministic sequence (γ_n) is nonincreasing, $0 < \gamma_n < 1$, and*

$$\sum_{n=1}^{\infty} \gamma_n = \infty, \qquad \sum_{n=1}^{\infty} \gamma_n^2 < \infty. \tag{4.6}$$

4.2 Regularity of the full observable

Lemma 4.1 (Polynomial and weighted Lipschitz bounds)**.** *For every compact K in the mean–covariance parameter domain, a constant A_K exists such that for all $\theta, \theta' \in K \times \mathrm{St}(d, r)$,*

$$\begin{aligned} \|H(\theta, x)\| &\leq A_K(1 + \|x\|^2), \\ \|H(\theta, x) - H(\theta', x)\| &\leq A_K(1 + \|x\|^2)\|\theta - \theta'\|. \end{aligned} \tag{4.7}$$

For the weight Equation 4.3, H is locally V^β-Lipschitz for any fixed $0 < \beta \leq 1/2$.

Proof. Write $z = x - \mu$, $z' = x - \mu'$. From $\|W\|_{\mathrm{op}} = 1$ and $\|I - WW^\top\|_{\mathrm{op}} \leq 1$,

$$\|(I - WW^\top)zz^\top W\|_F \leq \|z\|^2.$$

For parameter differences, use

$$zz^\top - z'z'^\top = (z - z')z^\top + z'(z - z')^\top,$$

together with $\|WW^\top - W'W'^\top\|_{\mathrm{op}} \le 2\|W - W'\|_F$. Add and subtract terms with μ' and W' one at a time. Because μ, μ' are bounded on K, every difference is bounded by $A_K(1 + \|x\|^2)\|\theta - \theta'\|$. The first two coordinates of H are simpler. Under (M), V^β dominates each fixed polynomial, so the weighted statements follow. □

4.3 Poisson equation and its parameter regularity

Put $f_\theta(x) = H(\theta, x) - h(\theta)$. The relevant Poisson solution is **Euclidean-valued**, not manifold-valued:

$$u_\theta - P_\theta u_\theta = f_\theta, \qquad u_\theta = \sum_{j=0}^{\infty} (P_\theta^j H_\theta - \pi H_\theta). \tag{4.8}$$

Existence alone is not enough: changing θ generates differences of $P_\theta u_\theta$, which must be controlled.

Lemma 4.2 (Uniform bounds and Lipschitz regularity)**.** *On every compact $K \times \mathrm{St}(d, r)$, there exists L_K such that*

$$\begin{aligned} \|u_\theta\|_{V^\beta} + \|P_\theta u_\theta\|_{V^\beta} &\le L_K, \\ \|u_\theta - u_{\theta'}\|_{V^\beta} + \|P_\theta u_\theta - P_{\theta'} u_{\theta'}\|_{V^\beta} &\le L_K \|\theta - \theta'\|. \end{aligned} \tag{4.9}$$

Proof. Apply Proposition 3, equations (18)–(19), of [1] to each coordinate of f_θ, using Assumption 4.2 and Lemma 4.1. The common invariant law and weighted geometric ergodicity ensure convergence of the Poisson series. The operator regularity A2 is used for the Lipschitz estimate, as is explicit in A&M's appendix proof. As the observable takes values in a fixed finite-dimensional ambient space, coordinatewise estimates suffice. □

5 Euclidean reduction and accumulated errors

There are two different perturbations: an error caused by replacing the Stiefel retraction by its first-order Euclidean expansion, and the dependence of successive Monte Carlo observations. Treating them separately makes the proof shorter and more transparent.

5.1 A global second-order retraction estimate

Lemma 5.1 (Polar retraction bounds)**.** *For every horizontal tangent vector U at $W \in \mathrm{St}(d, r)$, constants $c_r, c_r' < \infty$, depending only on r, satisfy*

$$\|R_W(U) - W - U\|_F \le c_r \|U\|_F^2, \qquad \|R_W(U) - W\|_F \le c_r' \|U\|_F. \tag{5.1}$$

Proof. Since $W^\top U = 0$, we have $(W + U)^\top (W + U) = I_r + U^\top U$, so

$$R_W(U) - W - U = (W + U)[(I_r + U^\top U)^{-1/2} - I_r].$$

When $\|U\|_F \le 1$, the scalar inequality $0 \le 1-(1+t)^{-1/2} \le t/2$, applied to the eigenvalues of $U^\top U$, yields $\|(I_r+U^\top U)^{-1/2} - I_r\|_F \le \|U\|_F^2/2$. Also $\|W+U\|_{\mathrm{op}} \le \sqrt{2}$. Thus the quadratic bound holds uniformly for small increments. For $\|U\|_F > 1$, the triangle inequality and $\|R_W(U)\|_F = \|W\|_F = \sqrt{r}$ give

$$\|R_W(U) - W - U\|_F \le 2\sqrt{r} + \|U\|_F \le (2\sqrt{r}+1)\|U\|_F^2.$$

The linear estimate follows for small U from the quadratic one, and for large U from $\|R_W(U) - W\|_F \le 2\sqrt{r}$. This explicit global bound is convenient with unbounded MCMC observations; the local Taylor principle is described in [2], Section 4.1. □

Writing $G_n = G(\mu_n, W_n, X_{n+1})$, Equation 3.5 becomes

$$W_{n+1} = W_n + \gamma_{n+1} G_n + r_{n+1}, \qquad \|r_{n+1}\|_F \le c_r \gamma_{n+1}^2 \|G_n\|_F^2. \tag{5.2}$$

Lemma 5.2 (Retraction errors are summable on eventually compact trajectories)**.** *Under Assumption 4.2, Assumption 4.3 and Lemma 4.1, on the event that (μ_n, Γ_n) eventually stays in some compact subset of $\mathbb{R}^d \times \mathbb{S}^d_{++}$,*

$$\sum_{n=0}^{\infty} \|r_{n+1}\|_F < \infty \quad \textit{almost surely.} \tag{5.3}$$

Proof. For a recursion stopped at its first exit from a compact K, Equation 5.2 and Equation 4.7 give $\|r_{n+1}\|_F \le C_K \gamma_{n+1}^2 V^{2\beta}(X_{n+1})$. As $2\beta \le 1$, the localized moment bound Equation 4.5 makes the expectation of the nonnegative series at most $C_K \sum_n \gamma_{n+1}^2$, which is finite. Exhaust the open mean–covariance domain by countably many compacts; an eventually confined trajectory lies in a stopped regime after a finite prefix. □

5.2 Decompose the controlled Markov fluctuation

The ambient recursion is

$$\theta_{n+1} = \theta_n + \gamma_{n+1}[h(\theta_n) + \xi_{n+1}] + (0, 0, r_{n+1}), \quad \xi_{n+1} = H(\theta_n, X_{n+1}) - h(\theta_n). \tag{5.4}$$

Let u_θ solve Equation 4.8, and write $v_\theta = P_\theta u_\theta$. For $k \ge 1$, define

$$M_k = u_{\theta_{k-1}}(X_k) - v_{\theta_{k-1}}(X_{k-1}). \tag{5.5}$$

Because X_k is conditionally distributed according to $P_{\theta_{k-1}}$, we have $\mathbb{E}[M_k \mid \mathcal{F}_{k-1}] = 0$. Subtracting this quantity from the Poisson identity gives

$$\xi_k = M_k + v_{\theta_{k-1}}(X_{k-1}) - v_{\theta_{k-1}}(X_k).$$

Multiplying by the step sizes and summing by parts, for $m \ge n \ge 1$,

$$\begin{aligned}\sum_{k=n}^{m}\gamma_k\xi_k = \sum_{k=n}^{m}\gamma_k M_k \\ &+ \gamma_n v_{\theta_{n-1}}(X_{n-1}) - \gamma_m v_{\theta_{m-1}}(X_m) \\ &+ \sum_{k=n}^{m-1}(\gamma_{k+1}-\gamma_k)v_{\theta_k}(X_k) \\ &+ \sum_{k=n}^{m-1}\gamma_k[v_{\theta_k}(X_k) - v_{\theta_{k-1}}(X_k)].\end{aligned} \tag{5.6}$$

The decomposition has four conceptually distinct pieces: a martingale, two endpoints, step-size variation, and **accumulated differences of Poisson solutions**. In particular, it is not a theorem about arbitrary retraction errors: those were already controlled in Lemma 5.2.

Lemma 5.3 (Weighted noise has vanishing tails)**.** *Under the assumptions of Section 4, on eventually compact parameter trajectories,*

$$\lim_{n\to\infty}\sup_{m\geq n}\left\|\sum_{k=n}^{m}\gamma_k[H(\theta_{k-1},X_k)-h(\theta_{k-1})]\right\| = 0 \quad \textit{almost surely.} \tag{5.7}$$

Proof. The estimates are detailed in Section 9. In summary, the first line of Equation 5.6 converges by the martingale L^2 theorem and $\sum\gamma_k^2<\infty$. Its endpoints vanish by Borel–Cantelli. The third line has absolutely summable coefficients because γ_k is nonincreasing. For the final line, combine Poisson Lipschitz regularity with the parameter increment estimate

$$\|\theta_k-\theta_{k-1}\| \leq C_K\gamma_k V^\beta(X_k), \tag{5.8}$$

which follows from Lemma 5.1 and the first two Euclidean recursions. Hence the last line is bounded in absolute value by $C_K\sum_k\gamma_k^2V^{2\beta}(X_k)$, finite almost surely on the localized event. □

5.3 ODE tracking, without a manifold-noise theorem

Set $t_n=\sum_{k=1}^{n}\gamma_k$, and linearly interpolate the ambient coordinates between t_n and t_{n+1}. For any fixed $T>0$,

$$\sup_{\substack{m\geq n\\ t_m-t_n\leq T}}\left\|\sum_{k=n}^{m-1}[\gamma_{k+1}\xi_{k+1}+(0,0,r_{k+1})]\right\|\to 0. \tag{5.9}$$

The averaged field is locally Lipschitz; its Euler error on a compact set is $O(\sum\gamma_k^2)$. Comparing the interpolation with the integral equation for Equation 3.9 and applying Grönwall's inequality proves that the interpolation is an **asymptotic pseudotrajectory** of the ODE. Its values between consecutive Stiefel iterates need not lie on the manifold; their distance to the manifold vanishes because Equation 5.8 and the weighted moment bound imply the step increment tends to zero. The limiting flow itself preserves Stiefel orthogonality.

This is the usual Euclidean stochastic approximation method [4], with the extra geometric term shown explicitly.

6 Lyapunov structure of the averaged flow

The mean and covariance estimates, and the Oja frame, are driven by the *same centred observations*. This suggests choosing a Lyapunov function whose geometric component is centred at the evolving mean μ, rather than regarding the dependence of W on μ as an external forcing. The natural choice is the expected residual of orthogonal projection onto the subspace spanned by W. Its derivative has a useful exact sign along the full averaged flow.

6.1 Mean–covariance dissipation

Let $v = \mu - m$, $Q(\mu) = C + vv^\top$, and $A = \Gamma^{-1}$. As in [1], equation (52), define

$$w_{\mathrm{AM}}(\mu, \Gamma) = \log\det\Gamma + v^\top \Gamma^{-1} v + \operatorname{tr}(\Gamma^{-1} C). \tag{6.1}$$

Lemma 6.1 (Exact mean–covariance dissipation)**.** *Along Equation 3.9,*

$$\frac{\mathrm{d}}{\mathrm{d}t} w_{\mathrm{AM}}(\mu, \Gamma) = -D_{\mathrm{AM}}(\mu, \Gamma),$$

where

$$D_{\mathrm{AM}} = 2v^\top \Gamma^{-1} v + \left\| \Gamma^{-1/2} (\Gamma - Q(\mu)) \Gamma^{-1/2} \right\|_F^2 \geq 0. \tag{6.2}$$

Moreover, $D_{\mathrm{AM}} = 0$ if and only if $(\mu, \Gamma) = (m, C)$.

Proof. Differentiate the two parameter coordinates independently. The identities

$$\mathrm{d}(\log\det\Gamma)[U] = \operatorname{tr}(\Gamma^{-1} U), \qquad \mathrm{d}(\Gamma^{-1})[U] = -\Gamma^{-1} U \Gamma^{-1}$$

give

$$\nabla_\mu w_{\mathrm{AM}} = 2Av, \qquad \nabla_\Gamma w_{\mathrm{AM}} = A - AQA = A(\Gamma - Q)A.$$

The corresponding averaged dynamics are $\dot\mu = -v$ and $\dot\Gamma = Q - \Gamma$. Thus

$$\begin{aligned} \dot w_{\mathrm{AM}} &= \langle 2Av, -v \rangle + \operatorname{tr}\{A(\Gamma - Q)A(Q - \Gamma)\} \\ &= -2v^\top Av - \operatorname{tr}\{A(\Gamma - Q)A(\Gamma - Q)\}. \end{aligned}$$

As A is positive definite and $\Gamma - Q$ is symmetric, the trace is $\|A^{1/2}(\Gamma - Q)A^{1/2}\|_F^2$. Both nonnegative quantities vanish exactly when $v = 0$ and $\Gamma = Q = C$. □

6.2 Compactness of the mean–covariance sublevel sets

The coercivity of w_{AM} will also control the noncompact coordinates of our joint Lyapunov function.

Put $S = C^{-1/2} \Gamma C^{-1/2} \succ 0$. Then

$$w_{\mathrm{AM}} = \log\det C + \log\det S + \operatorname{tr}(S^{-1}) + v^\top \Gamma^{-1} v. \tag{6.3}$$

Writing $s_1, \dots, s_d > 0$ for the eigenvalues of S, the middle terms are $\sum_i (\log s_i + s_i^{-1})$. The scalar function $s \mapsto \log s + s^{-1}$ is bounded below by 1, and tends to $+\infty$ as $s \downarrow 0$ or $s \uparrow \infty$. Consequently, a sublevel bound on w_{AM} bounds all eigenvalues of S away from 0 and ∞. The remaining nonnegative quadratic term then bounds $\|v\|$.

In particular, each nonempty sublevel set

$$K_L = \{(\mu, \Gamma) : w_{\mathrm{AM}}(\mu, \Gamma) \le L\} \tag{6.4}$$

is compact **inside** $\mathbb{R}^d \times \mathbb{S}^d_{++}$, with constants $0 < a_L \le b_L < \infty$ and $M_L < \infty$ such that

$$a_L I \preceq \Gamma \preceq b_L I, \qquad \|\mu - m\| \le M_L \qquad \text{on } K_L. \tag{6.5}$$

The positive lower eigenvalue bound is part of compactness in the *open* positive-definite domain, not a consequence of bounded matrix entries alone.

6.3 The expected reconstruction error

For $W \in \mathrm{St}(d, r)$ write

$$P_W = I - WW^\top.$$

This is the orthogonal projector onto the complement of $\mathrm{span}(W)$. The squared length of the residual after projecting a centred observation $x - \mu$ onto $\mathrm{span}(W)$ is $\|P_W(x-\mu)\|^2$. Averaging over the common invariant target law gives the natural PCA objective

$$\begin{aligned} \mathcal{R}(\mu, W) &= \frac{1}{2}\pi[\|P_W(X-\mu)\|^2] \\ &= \frac{1}{2}\operatorname{tr}(P_W Q(\mu)) \\ &= \frac{1}{2}\operatorname{tr}(P_W C) + \frac{1}{2} v^\top P_W v. \end{aligned} \tag{6.6}$$

The second equality uses Equation 3.7. Notice that $\mathcal{R} \ge 0$ and, importantly, the centring parameter μ is the **same** parameter used in both the covariance and the sample-driven Oja recursions. This choice connects the objective directly to the algorithm and to classical reconstruction-error Lyapunov arguments for PCA [3].

Lemma 6.2 (The Oja field is the negative reconstruction-error gradient)**.** *For fixed μ, the Riemannian gradient on the embedded Stiefel manifold is*

$$\operatorname{grad}_W \mathcal{R}(\mu, W) = -P_W Q(\mu) W. \tag{6.7}$$

For fixed W, the Euclidean gradient in the mean coordinate is

$$\nabla_\mu \mathcal{R}(\mu, W) = P_W v. \tag{6.8}$$

Hence, along the coupled mean-field system Equation 3.9,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{R}(\mu, W) = -\|P_W v\|^2 - \|P_W Q(\mu) W\|_F^2.} \tag{6.9}$$

Proof. Because P_W is a symmetric idempotent matrix, $v^\top P_W v = \|P_W v\|^2$. Differentiating the last expression in Equation 6.6 with respect to μ immediately gives $\nabla_\mu \mathcal{R} = P_W v$.

For the W derivative, hold μ fixed and use $\operatorname{tr}(P_W Q) = \operatorname{tr} Q - \operatorname{tr}(W^\top Q W)$. The first term does not depend on W. Since Q is symmetric,

$$\mathrm{d}_W \mathcal{R}[U] = -\operatorname{tr}(U^\top Q W) = -\langle QW, U \rangle_F.$$

Thus the ambient Euclidean gradient is $-QW$. Its orthogonal projection onto the Stiefel tangent space is

$$-QW + W\ \operatorname{sym}(W^\top Q W) = -(I - WW^\top)QW = -P_W Q W,$$

where $\operatorname{sym}(W^\top Q W) = W^\top Q W$ by symmetry of Q.

Finally, the first and third components of Equation 3.9 are

$$\dot{\mu} = -v, \qquad \dot{W} = P_W Q(\mu) W = -\operatorname{grad}_W \mathcal{R}.$$

Apply the chain rule on $\mathbb{R}^d \times \mathrm{St}(d, r)$:

$$\begin{aligned} \dot{\mathcal{R}} &= \langle P_W v, -v \rangle + \langle -P_W Q W, P_W Q W \rangle_F \\ &= -\|P_W v\|^2 - \|P_W Q W\|_F^2. \end{aligned}$$

The first negative square is the contribution of the evolving mean; the second is the dissipation generated by the Oja step. □

6.4 A Lyapunov function for the full coupled system

Proposition 6.1 (Global joint Lyapunov dissipation)**.** *For every $\lambda > 0$ define*

$$\mathcal{L}_\lambda(\mu, \Gamma, W) = w_{\mathrm{AM}}(\mu, \Gamma) + \lambda \mathcal{R}(\mu, W). \tag{6.10}$$

Along every solution of Equation 3.9 in $\mathbb{R}^d \times \mathbb{S}^d_{++} \times \mathrm{St}(d, r)$,

$$\boxed{\begin{aligned} \dot{\mathcal{L}}_\lambda = &- D_{\mathrm{AM}}(\mu, \Gamma) \\ &- \lambda \|P_W(\mu - m)\|^2 \\ &- \lambda \|P_W Q(\mu) W\|_F^2 \le 0. \end{aligned}} \tag{6.11}$$

Equality holds precisely on

$$\mathcal{E} = \{(m, C, W) : (I - WW^\top) C W = 0\}. \tag{6.12}$$

Every nonempty sublevel set of $\mathcal{L}_\lambda$ is compact and forward invariant under Equation 3.9.

Proof. Add the identities Equation 6.2 and Equation 6.9; there is no unspecified remainder. All three terms on the right-hand side are nonpositive for every $\lambda > 0$.

If $\dot{\mathcal{L}}_\lambda = 0$, then $D_{\mathrm{AM}} = 0$, so $\mu = m$ and $\Gamma = C$ by Lemma 6.1. It follows that $Q(\mu) = C$. The final square vanishes exactly when $P_W C W = 0$. Conversely every such point makes all three terms vanish.

For compactness, $\mathcal{R} \geq 0$ implies

$$\{\mathcal{L}_\lambda \leq L\} \subseteq K_L \times \mathrm{St}(d, r).$$

The right-hand side is compact by Equation 6.4 and compactness of the Stiefel manifold. The left-hand side is closed there by continuity of $\mathcal{L}_\lambda$, hence compact. The derivative inequality shows that its value cannot increase along an ODE solution; therefore the sublevel set is forward invariant. □

The proposition is **global for the averaged ODE**: its sign does not depend on restricting the drift calculation to a compact set. This does not, by itself, establish pathwise boundedness of the *stochastic* recursion. Uniform kernel and Poisson estimates, and stochastic stability or reinitialization, remain necessary for an unconditional stochastic convergence theorem.

6.5 Equilibria and Lyapunov values

The condition $P_W C W = 0$ is equivalent to

$$CW = WW^\top CW. \tag{6.13}$$

Thus $\mathrm{span}(W)$ is an invariant r-dimensional subspace of the symmetric matrix C. The equilibrium set need not select the leading subspace.

For later use, observe that at a joint equilibrium,

$$\mathcal{L}_\lambda(m, C, W) = \log\det C + d + \frac{\lambda}{2}(\operatorname{tr} C - \operatorname{tr}(W^\top CW)). \tag{6.14}$$

The trace $\operatorname{tr}(W^\top CW)$ is a sum of r eigenvalues of C, counted with multiplicity. Hence $\mathcal{L}_\lambda$ assumes only **finitely many distinct values** on $\mathcal{E}$, even when eigenspaces have multiplicity and contain infinitely many possible frames. This property will allow us to apply the internally chain-transitive limit-set argument of [4].

7 Convergence theorem and its scope

We now combine the stochastic averaging, retraction and Lyapunov arguments. It is important to distinguish the localized theorem below from a global theorem for untruncated adaptation.

For a compact set $K \subset \mathbb{R}^d \times \mathbb{S}^d_{++}$, define the event

$$\Omega_K = \{\exists N < \infty : (\mu_n, \Gamma_n) \in K \text{ for all } n \geq N\}, \tag{7.1}$$

and let Ω_{ec} be the union of these events over a fixed countable compact exhaustion of the mean–covariance parameter domain.

Theorem 7.1 (Localized convergence of adaptive Markov–Oja)**.** *Suppose the target satisfies Assumption 4.1 and the Gaussian random-walk Metropolis family satisfies Assumption 4.2. Let $(X_n, \mu_n, \Gamma_n, W_n)$ follow Definition 3.1 with Assumption 4.3, $\Gamma_0 \succ 0$, $W_0 \in \mathrm{St}(d,r)$, and $\mathbb{E}V(X_0) < \infty$. Then, almost surely on Ω_{ec},*

$$\boxed{\mu_n \longrightarrow m, \qquad \Gamma_n \longrightarrow C, \qquad \mathrm{dist}_F(W_n, \mathcal{E}_C) \longrightarrow 0,} \tag{7.2}$$

where

$$\mathcal{E}_C = \{W \in \mathrm{St}(d,r) : (I - WW^\top)CW = 0\}. \tag{7.3}$$

In particular this theorem does not assert convergence to the dominant principal subspace, nor almost-sure eventual compactness of the untruncated recursion.

Proof. **1. A Euclidean representation of the actual recursion.** By Lemma 5.1 the iterates satisfy Equation 5.4 with a quadratic remainder. Under the localized weighted moment bounds, Lemma 5.2 makes its accumulated tails vanish.

2. Markov noise. The Poisson equation Equation 4.8, its Lipschitz property Lemma 4.2, and the four-term identity Equation 5.6 show through Lemma 5.3 that the weighted noise has vanishing tails. This does not assume that the accepted states are independent or stationary at any finite time.

3. ODE tracking and compactness. On Ω_{ec} the interpolation is precompact and is an asymptotic pseudotrajectory of Equation 3.9 by Equation 5.9. Invariance of the Stiefel manifold for the limiting ODE means that the limit set lies in $\mathbb{R}^d \times \mathbb{S}^d_{++} \times \mathrm{St}(d,r)$. By Theorem 5.7 of [4], its limit set is internally chain transitive.

4. A strict Lyapunov function around the limit set. Fix an eventual compact set K. Choose any $\lambda > 0$ and then an L large enough that $K \times \mathrm{St}(d,r)$ is contained in the joint sublevel set $\{\mathcal{L}_\lambda \le L\}$. By Proposition 6.1 this is a compact, forward-invariant set for the limiting ODE. Its Lyapunov function $\mathcal{L}_\lambda$ has strictly negative derivative away from $\mathcal{E}$, and its values on $\mathcal{E}$ form a finite set.

5. Identify the stochastic limit set. Proposition 6.4 of [4] states that an internally chain-transitive set for a compact flow with a strict Lyapunov function must lie in the zero-dissipation set when the latter has Lyapunov values with empty interior. The set here is $\mathcal{E}$, and finiteness of $\mathcal{L}_\lambda(\mathcal{E})$ verifies the condition. Every limit point of the interpolated trajectory therefore belongs to $\mathcal{E}$. Because the trajectory is precompact, its distance to $\mathcal{E}$ tends to zero. Reading off the three coordinates gives Equation 7.2. □

7.1 How to read this statement

It is a convergence theorem for the un-reprojected recursion **conditional on eventual compactness**. The qualification is mathematically precise: for any path on which the estimates stay eventually inside a compact subset of the positive-definite domain, the stochastic and geometric perturbations have negligible tails and the limit-set argument applies.

A&M achieve unconditional stability for their mean–covariance recursion with a random-reinitialization mechanism, under an open Euclidean parameter-domain formulation. The extra Stiefel coordinate is compact, and its retraction contributes a summable error. These

facts strongly suggest adapting that stability proof, but Theorem 7.1 does not silently equate a summable perturbation with an automatic invocation of A&M's Theorem 11.

Corollary 7.1 (Finite-reinitialization consequence)**.** *Suppose an implementation of the enlarged recursion uses an A&M-type expanding-compact reinitialization mechanism, and suppose its number of reinitializations is almost surely finite and its post-last-reinitialization trajectory is eventually contained in a compact mean–covariance set. Then Equation 7.2 holds almost surely.*

Proof. After the last reinitialization the dynamics have the form Definition 3.1, possibly with a shifted step-size index. A finite prefix does not affect any summability, ODE-time or limit-set argument; apply Theorem 7.1 to that tail. □

The premise that the enlarged algorithm has finitely many reinitializations is an **additional stability obligation**. A direct manifold-compatible adaptation of A&M's Theorem 11 would close the unconditional version; it is not proved in this draft.

7.2 A deterministic check on the limiting flow

The first equation of Equation 3.9 is explicit:

$$v(t) = e^{-t}v(0).$$

Setting $D(t) = \Gamma(t) - C$, the second becomes $\dot{D} = vv^\top - D$, so

$$D(t) = e^{-t}D(0) + (e^{-t} - e^{-2t})v(0)v(0)^\top. \tag{7.4}$$

Thus $(\mu(t), \Gamma(t)) \to (m, C)$ exponentially. The Oja equation is therefore an asymptotically autonomous perturbation of $\dot{W} = B_C(W)$. This elementary calculation independently supports the equilibrium picture established by the composite Lyapunov argument, but it is not a substitute for the stochastic Markov-noise analysis.

8 Interpretation and further questions

The analysis establishes a sharp division of responsibilities. All frozen Metropolis kernels target the same π, so their common invariant distribution determines the deterministic Oja mean field. A&M's Markov theory controls the failure of successive observations to be independent. The polar retraction preserves orthogonality while contributing only an absolutely summable second-order error. The mean–covariance energy controls noncompact coordinates, while the expected reconstruction error has an exact decreasing derivative in both the evolving mean and the Oja frame. Their sum identifies the joint equilibrium set.

The result should not be read as an assertion that online Oja necessarily selects the leading r-dimensional eigenspace. Its limiting stationary set contains **every invariant r-subspace** of C, including non-leading eigenspaces. Deterministically, the leading subspace is locally attracting under an eigengap, whereas non-leading invariant subspaces have unstable directions; showing that Markov-driven stochastic iterates almost surely avoid unstable equilibria requires additional hypotheses on noise excitation and is a distinct problem. The Lyapunov literature of [3] is relevant to that refinement.

There is also an important distinction between two possible algorithms. We have analyzed a *sample-driven* Oja increment, using $(X_{n+1} - \mu_n)(X_{n+1} - \mu_n)^\top$. Its mean-field perturbation from the target Oja flow depends on $(\mu - m)(\mu - m)^\top$. If one instead updates W using the *estimated covariance* Γ_n, then $\dot{W} = (I - WW^\top)\Gamma W$, with a different coupling error depending on $\Gamma - C$. Its convergence analysis requires an objective and noise decomposition consistent with that distinct update, and is not claimed by the present theorem.

Finally, the localized result does not settle the probability-one finite-reinitialization question for the **enlarged** algorithm. Under the concrete Gaussian random-walk kernel, the necessary kernel bounds are available from A&M, and the new coordinate is compact. Establishing the exact manifold-compatible stability theorem, with the reinitialization map and boundary conditions stated, is the remaining step before claiming unconditional convergence of the original adaptive procedure. This is recorded transparently in the internal proof audit.

The present theorem is therefore a useful foundation for investigating principal-subspace adaptation with Markov rather than independent innovations. It identifies the precise non-i.i.d. bookkeeping, proves a localized limiting-subspace result, and leaves the stronger global stability and leading-eigenspace selection questions clearly separated.

9 Detailed weighted Poisson estimates

For completeness, we account for every line in Equation 5.6. Work first up to the exit time τ_K from a compact parameter region. By the locally uniform geometric drift and minorization properties, and after enlarging K slightly if necessary, stopped expectations of $V(X_k)$ are uniformly bounded. The same bounds apply to $V^{2\beta}(X_k)$, since $2\beta \le 1$.

Martingale. From Equation 4.9 and conditional Jensen's inequality,

$$\mathbb{E}\left[\|M_k\|^2 \mathbf{1}_{\{\tau_K \ge k\}}\right] \le C_K\, \mathbb{E}[V^{2\beta}(X_{k-1}) + V^{2\beta}(X_k); \tau_K \ge k] \le C_K. \tag{9.1}$$

Thus $\sum_k \gamma_k M_k \mathbf{1}_{\{\tau_K \ge k\}}$ is an L^2-bounded martingale, and the martingale convergence theorem makes its tails vanish uniformly over their terminal index.

Endpoints. Markov's inequality and Equation 4.9 give

$$\sum_k \mathbb{P}\left(\gamma_k \|v_{\theta_{k-1}}(X_{k-1})\| > \varepsilon,\, \tau_K \ge k\right) \le \varepsilon^{-2} C_K \sum_k \gamma_k^2 < \infty. \tag{9.2}$$

Hence $\gamma_k v_{\theta_{k-1}}(X_{k-1}) \to 0$ almost surely on the localized event, by Borel–Cantelli. The two endpoint contributions therefore vanish even when the upper index m varies.

Step-size variation. Because (γ_k) is nonincreasing and converges to zero,

$$\sum_{k=1}^{\infty} |\gamma_{k+1} - \gamma_k| = \gamma_1 < \infty.$$

Since $\mathbb{E}[V^\beta(X_k); \tau_K \ge k] \le C_K$, the expectation of the absolute sum $\sum |\gamma_{k+1} - \gamma_k| \|v_{\theta_k}(X_k)\|$ is finite. Thus that sum converges almost surely.

Parameter variation. Use the exact place where the parameter increments enter:

$$\begin{aligned} \gamma_k \|v_{\theta_k}(X_k) - v_{\theta_{k-1}}(X_k)\| &\le L_K \gamma_k \|\theta_k - \theta_{k-1}\| V^\beta(X_k) \\ &\le C_K \gamma_k^2 V^{2\beta}(X_k). \end{aligned} \tag{9.3}$$

The sum of the right-hand side has finite expectation on the stopped process. This is a bound for differences of the **Poisson solution evaluated at successive parameters**, rather than a bound for a general extra error in the SA recursion.

The four convergent expressions prove Equation 5.7 by passing to their tails. Taking a countable compact exhaustion and ignoring a finite initial segment transfers the localized claim to every eventually compact trajectory.

10 Supplementary geometric and deterministic details

10.1 The Stiefel tangent and invariance statements

The tangent space at $W \in \mathrm{St}(d, r)$ is

$$T_W \mathrm{St}(d, r) = \{U \in \mathbb{R}^{d \times r} : W^\top U + U^\top W = 0\}.$$

For $G(\mu, W, x)$ from Equation 3.3, $W^\top G = 0$. Hence G is not merely tangent: it is horizontal, with the particularly simple polar retraction Equation 3.4. Its image satisfies

$$R_W(U)^\top R_W(U) = (I + U^\top U)^{-1/2}(I + U^\top U)(I + U^\top U)^{-1/2} = I_r.$$

The averaged field has the same horizontal form, so any ODE solution starting from a Stiefel matrix remains on Stiefel. Indeed

$$\frac{\mathrm{d}}{\mathrm{d}t}(W^\top W) = W^\top \dot{W} + \dot{W}^\top W = 0.$$

This is why the limiting problem does not require extending the ODE away from the manifold to *define its actual solution*, even though ambient Euclidean coordinates are used for stochastic error estimates.

10.2 Positive definiteness of the covariance iterate

The first two recursions are exactly Euclidean. The covariance recursion is

$$\Gamma_{n+1} = (1 - \gamma_{n+1})\Gamma_n + \gamma_{n+1}(X_{n+1} - \mu_n)(X_{n+1} - \mu_n)^\top.$$

As $0 < \gamma_{n+1} < 1$ and the second term is positive semidefinite,

$$\Gamma_{n+1} \succeq (1 - \gamma_{n+1})\Gamma_n \succ 0.$$

Thus Γ_n stays positive definite at every **finite** iteration. This does not establish that its smallest eigenvalue stays bounded away from zero as $n \to \infty$. Such a uniform lower bound requires the eventual-compactness/stability argument; the distinction is central to Equation 6.5.

10.3 The reconstruction-error objective and its geometric interpretation

The residual projector $P_W = I - WW^\top$ is symmetric and idempotent. Therefore the expected reconstruction error in Equation 6.6 is equivalently

$$\mathcal{R}(\mu, W) = \frac{1}{2}(\operatorname{tr} Q(\mu) - \operatorname{tr}(W^\top Q(\mu) W)).$$

At fixed μ, minimizing $\mathcal{R}$ means maximizing the variance explained by W. On the Stiefel manifold its negative Riemannian gradient is exactly the averaged sample-driven Oja field; see Lemma 6.2. The moving centring parameter contributes the further nonpositive term $-\|P_W(\mu - m)\|^2$ to the derivative. Thus the composite functional Equation 6.10 is a direct sum of the mean–covariance energy and an objective actually decreased by the complete averaged recursion.

10.4 Invariant subspaces and finitely many energy values

Let $C = U\,\mathrm{diag}(\lambda_1, \dots, \lambda_d)U^\top$, counting eigenvalues with multiplicity. If $\mathrm{span}(W)$ is invariant under the symmetric C, it is a direct sum of subspaces of eigenspaces. The trace $\mathrm{tr}(W^\top CW)$ equals a sum of exactly r entries from this eigenvalue list. The number of possible sums is finite, although there can be infinitely many orthonormal bases representing the same invariant subspace, and even infinitely many invariant subspaces when eigenvalues repeat. Therefore $\mathcal{L}_\lambda(\mathcal{E})$ is a finite subset of $\mathbb{R}$, as required by [4], Proposition 6.4.

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